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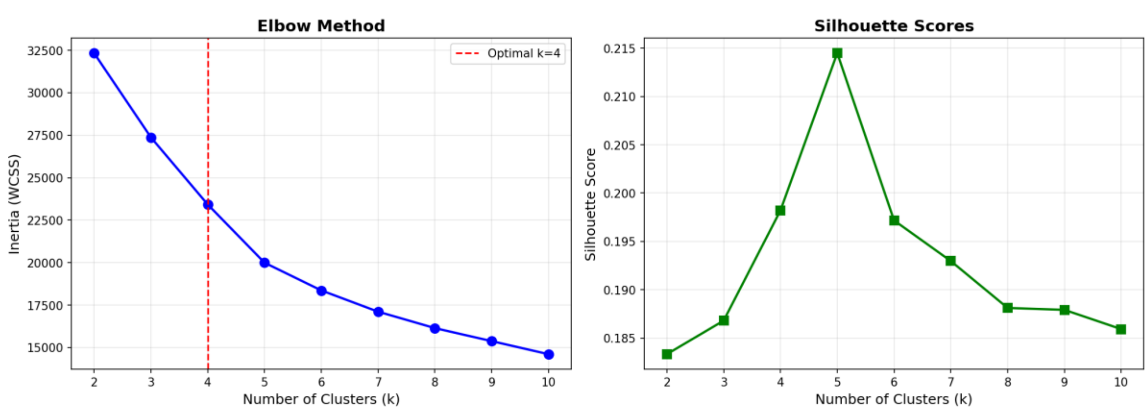
Part 4: Clustering - Student Segmentation & Risk Analysis

Objective

Segment students into meaningful groups using GPA, Attendance Percentage, Credits Completed, and Study Load. Techniques used include K-Means Clustering (with Elbow Method) and Hierarchical (Agglomerative) Clustering (with Dendrogram).

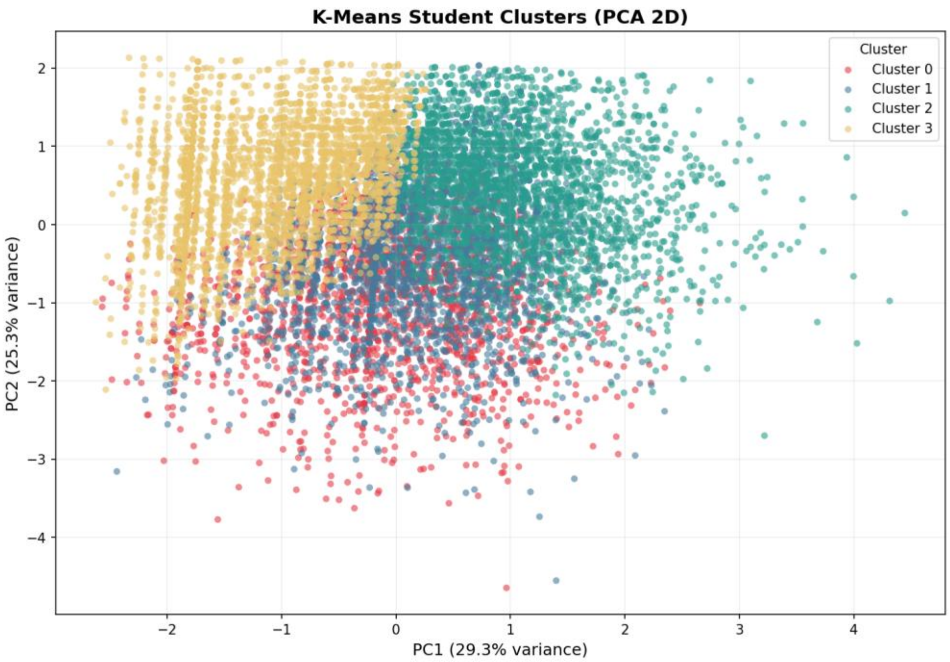
1. Data Preparation, Feature Scaling, and K-Means Elbow Method

```
features = ['GPA', 'Attendance_Pct', 'Credits_Completed', 'Study_Load']
X = df[features].values
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

inertias = []
silhouettes = []
k_range = range(2, 11)

for k in k_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_scaled)
    inertias.append(km.inertia_)
    silhouettes.append(silhouette_score(X_scaled, km.labels_))
```

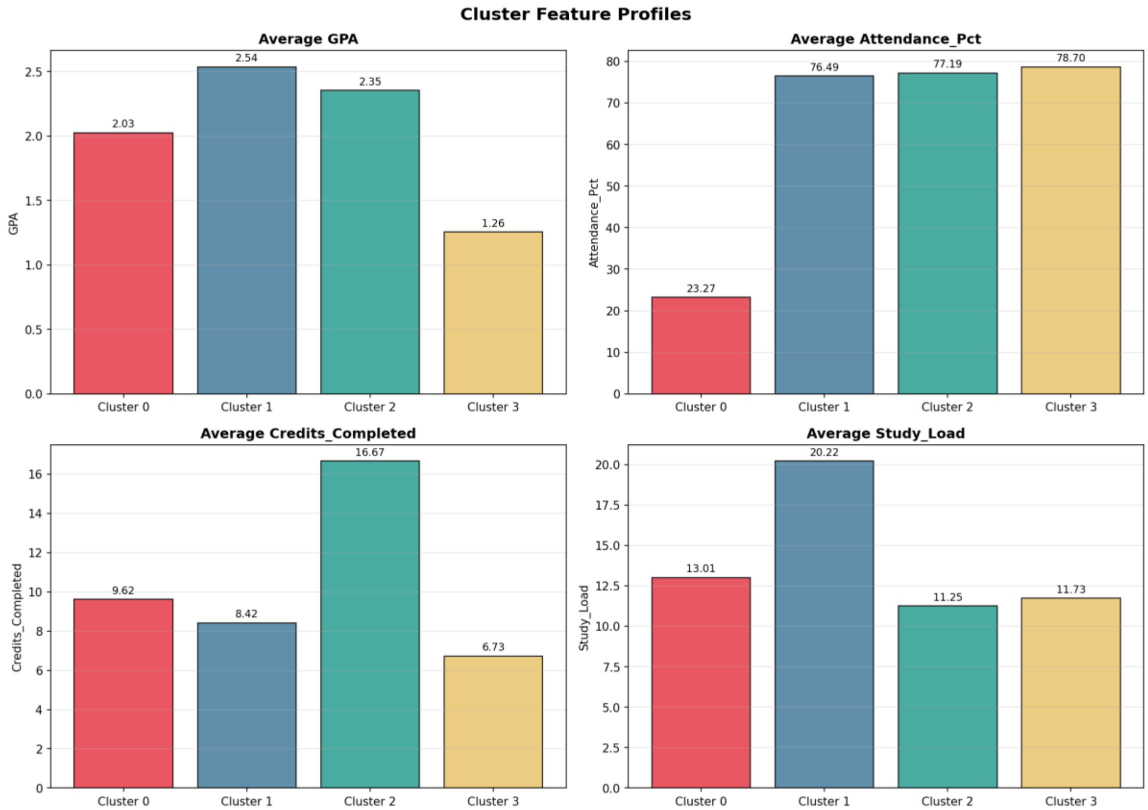


2. K-Means Clustering (k=4)

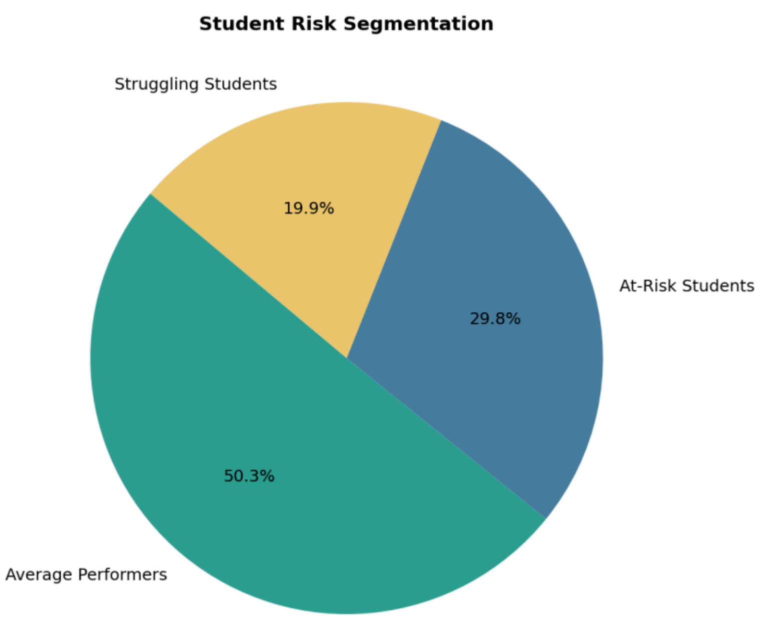
```
K = 4
kmeans = KMeans(n_clusters=K, random_state=42, n_init=10)
df['KMeans_Cluster'] = kmeans.fit_predict(X_scaled)

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)
df['PCA1'] = X_pca[:,0]
df['PCA2'] = X_pca[:,1]
```



3. Cluster Profile Analysis

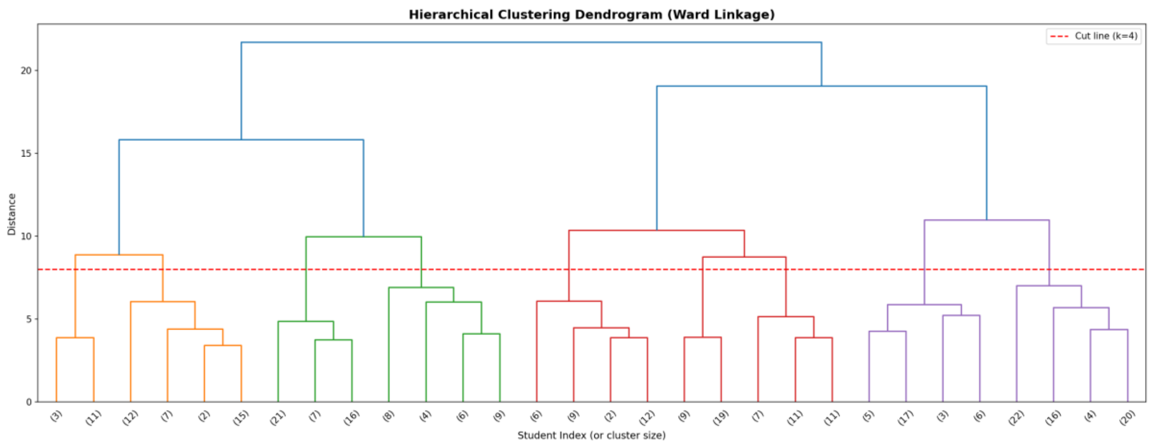


4. Risk Analysis - Segment Labels



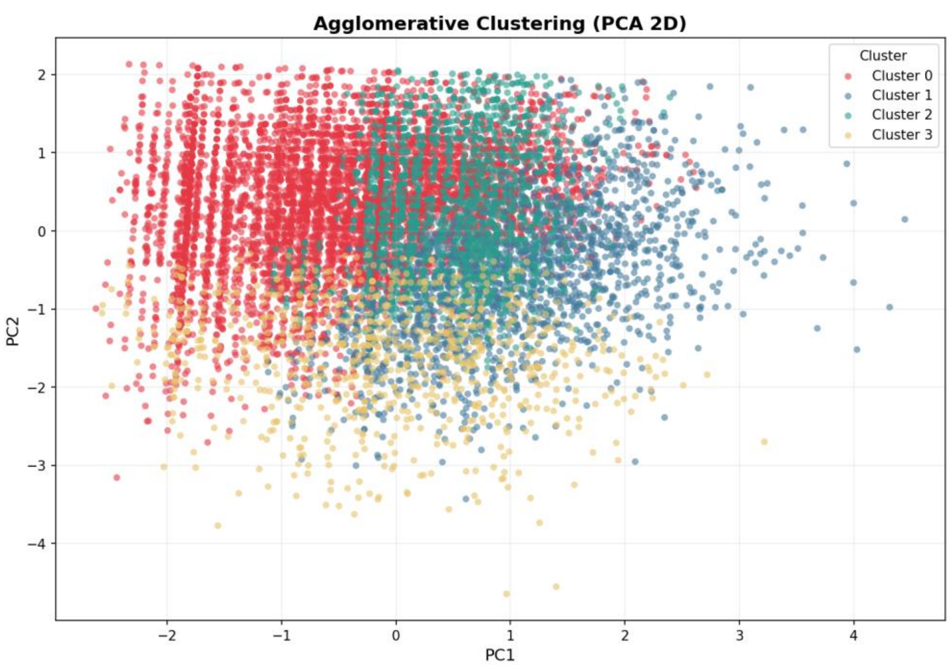
5. Hierarchical Clustering (Dendrogram)

```
sample_idx = np.random.choice(len(X_scaled), size=300, replace=False)
X_sample = X_scaled[sample_idx]
linked = linkage(X_sample, method='ward')
dendrogram(linked, truncate_mode='lastp', p=30)
```



6. Agglomerative Clustering

```
agg = AgglomerativeClustering(n_clusters=4, linkage='ward')
df['Agg_Cluster'] = agg.fit_predict(X_scaled)
```



Conclusion

Both K-Means and Agglomerative Clustering identified 4 optimal segments: High Achievers, Average Performers, Struggling Students, and At-Risk Students. This segmentation enables targeted interventions for students based on their behavioral and academic profiles.